



Clean and prepare data with help from AI

How many hours have you spent getting data ready for analysis, on things like cleaning spreadsheets, fixing bad data, changing data types to be accurate and representative, or one of my favorites, renaming columns? For me, the answer is, well, a whole lot of hours. In our field, clean data is the foundation for any analysis. As data analysts, we typically get large, incomplete, sometimes erroneous data sets, either from stakeholders, external vendors, or even just from old code and databases. For each data set, I can spend up to 30% of my time cleaning, validating, and preparing it for analysis. It can be tedious. AI has helped me reclaim some of these hours for deeper analysis to do the work I really love.

Gen AI Tools like Gemini can help data analysts identify data quality issues, standardize date formats, detect and remove duplicates, identify potential data set features, in record time. For example, Gemini can provide a detailed process for achieving a high quality data set. And with more advanced tools like Gemini for Google Workspace, we can grant access to a data set and have the tool complete the changes for us. Think of a recent data set you worked with. It could be a data set from this certificate program or from a project at work you recently completed. Pause the video if you have to, and really think of one. What were the most common issues you faced when trying to clean that data?

Bad data types, missing information, unclear field names? What if AI could help with all of that with a little prompting from you? Let me show you an example. Imagine I have a data set describing daily sales data. This data set has inconsistent date formats. Frustrating, right? How do I even go about fixing those?

Well, I can enter a prompt like this. I've received a sales data set that contains numerous formatting issues. Specifically, there are inconsistent date formats. I'm gonna go ahead and enter some generic dates as an example. Describe a step-by-step process for cleaning and standardizing the date formats exclusively in Google Sheets to ensure the dataset is ready for accurate analysis. Now, let's check out the output. Gemini provide steps on how to identify the columns of my data that may contain dates. Then, it offers tips to make it easier to figure out which fields have dates and how to pick a standard format so that all the dates are shown the same way. It even gives me notes on error handling and lets we know the safest way to convert the information into the standard format I choose. Pretty cool, right? Clean data for all. In this example, Gemini guided us through the process for cleaning and standardizing the date formats. But this is just the tip of the iceberg when it comes to gen AI's capabilities in data analysis. Many enterprise level AI tools can go even further by automating data cleaning tasks like these.

This can save valuable time and resources, allowing you to focus on things like strategic decision making. Now, try it out for yourself. Playing and experimenting with prompts is the best way to figure out how to get results that are useful for you. Open a data set you're currently working with and see how a gen AI tool can help. Getting the output you want can be a journey sometimes, and the only way to get there is to get comfortable, explore, and learn what path works for you.